

Harish Kumar S

Associate Software Developer

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<https://github.com/Harish-nika>

<https://harish-nika.github.io/>

PROFESSIONAL SUMMARY

Integrated MTech in Data Science (VIT, 2025). Associate Software Developer at FactEntry Data Solutions (A SIX Company) building production AI systems. Experienced in MLOps, Kubernetes, GitOps, RAG pipelines, computer vision, and scalable backend engineering. Delivered enterprise-grade AI platforms using FastAPI microservices, Redis-based distributed processing, and cloud-native deployment architectures.

- Built and deployed a production bond-extraction platform on K3s, leveraging 80+ CPU cores with Redis Streams and human-in-the-loop validation workflows.
- Implemented end-to-end GitOps delivery pipelines using GitLab CI, Helm, and Argo CD for automated staging deployments.
- Developed RAG-powered document intelligence systems using FAISS, Groq LLaMA, Streamlit, and FastAPI.
- Published the language-processing-tool package on PyPI for reusable NLP workflows.
- Trained and deployed YOLOv11 computer vision models on Roboflow for dental image analysis applications.

CORE COMPETENCIES

AI/ML Engineer | Machine Learning | Deep Learning | Computer Vision | NLP | RAG | Python | FastAPI | Kubernetes | K3s | Docker | Redis | PostgreSQL | GitLab CI | Argo CD | Helm | GitOps | MLOps | Roboflow | YOLO | FAISS | LangChain | Streamlit

EXPERIENCE

Associate Software Developer

FactEntry Data Solutions - A SIX Company | Dec 2025 - Present

- Shipped Data Mining Engine (DME) modules on K3s stage: corporate/municipal bond extraction, validation UI, and cluster operations portal.
- Implemented Redis-stream job dispatch with multi-node worker throughput and GitOps delivery (GitLab CI image build -> Helm chart promote -> Argo CD).
- Built FastAPI microservices, live WebSocket job telemetry, role-based CPU tiering, and domain-restricted enterprise auth.

MLOps and Data Science Intern

FactEntry Data Solutions - A SIX Company | Nov 2024 - May 2025

- Published language-processing-tool on PyPI (PDF language detection, OCR, document clustering) as self-driven internship contribution.
- Built Fixed Income Expert RAG system (FAISS, Groq LLaMA 3, Streamlit/FastAPI) as personal contribution during internship.

Associate Software Engineer Trainee Intern

MIMASOFT Technologies | Jun 2025 - Oct 2025

- Developed Dental ML Project: Roboflow YOLOv11 tooth and panoramic X-ray models with inference CLI.
- Contributed to backend integration, API workflows, and production-oriented software development practices.

EDUCATION

Integrated MTech - CSE (Data Science), Vellore Institute of Technology (2020 - 2025) - CGPA 7.8

Class XII, THIRU G V C Higher Secondary School (2020) - 79%

Class X, St Joseph Matriculation Higher Secondary School (2018) - 87.8%

TECHNICAL SKILLS

AI/ML: Machine Learning, Deep Learning, Generative AI, Computer Vision, NLP, RAG, FAISS, Sentence Transformers, Ollama, Roboflow, YOLOv11, Supervision

Engineering: Python, FastAPI, Streamlit, SQL, Docker, Kubernetes (K3s), Redis, PostgreSQL, LangChain, PyMuPDF, OpenCV

DevOps/GitOps: GitHub, GitLab CI, Argo CD, Helm, systemd, GitHub Actions, GitOps

PROJECTS

Data Mining Engine (DME)

Where: FactEntry Data Solutions (A SIX Company) - Associate Software Developer

Enterprise bond extraction on K3s - UI, API, Redis queue workers, GitOps CI/CD, validation workspace.

Tech: K3s, FastAPI, Redis, PostgreSQL, Argo CD

<https://harish-nika.github.io/dme-project.html>

Recruiting Agent - AI-Powered ATS

Where: Independent R&D - Agents Monorepo

Multi-step LLM ATS - resume OCR, ChromaDB JD matching, verification, suspicion scoring, and HR feedback loop.

Tech: FastAPI, React, SQLite, ChromaDB, Ollama, Groq

<https://harish-nika.github.io/recruiting-agent.html>

<https://github.com/Harish-nika/Agents>

Fixed Income Expert System

Where: FactEntry internship - personal contribution

RAG chatbot for financial bond PDFs - upload, embed, retrieve, LLaMA 3 answers via Groq.

Tech: FastAPI, Streamlit, FAISS, Groq, SBERT

<https://harish-nika.github.io/fexpert.html>

<https://github.com/Harish-nika/FExpert>

Cybersecurity Content Moderator

Where: VIT - Capstone project (academic)

Multimodal hate-speech and harmful content classification (text, image, PDF).

Tech: LLM, Streamlit, Docker, FastAPI

<https://harish-nika.github.io/content-moderator.html>

<https://github.com/Harish-nika/Content-moderator-image>

Language Distribution & Document Clustering Tool

Where: FactEntry internship - self contribution

PyPI-published PDF language detector, OCR, and clustering pipeline (v0.2.7).

Tech: Python, Tesseract, LangDetect, PyPI

<https://harish-nika.github.io/lang-tool.html>

https://github.com/Harish-nika/language_processing_tool

MedBot (Medical RAG Assistant)

Where: VIT - J Component, Soft Computing course

Medical PDF Q&A with SBERT + FAISS retrieval and Groq LLaMA 3 grounded responses.

Tech: Streamlit, FAISS, Sentence Transformers, Groq

<https://harish-nika.github.io/medbot.html>

<https://github.com/Harish-nika/medbot>

Dental ML Project

Where: MIMASOFT Technologies - trainee internship

YOLOv11 tooth and panoramic X-ray detection - Roboflow fine-tuning and inference CLI.

Tech: Roboflow, YOLOv11, inference-sdk, Supervision

<https://harish-nika.github.io/dental-project.html>

https://github.com/Harish-nika/Dental_ML_project

Disease Prediction using Ensemble Learning

Where: SmartInternz / Smart Bridge - Applied Data Science externship

Ensemble ML model for clinical trial disease prediction with MongoDB Atlas pipeline.

Tech: Python, Random Forest, SVM, MongoDB

<mailto:harishkumar56278@gmail.com?subject=Disease%20Prediction%20Project>

Real-Time Room Temperature & Humidity Visualization (AR)

Where: VIT - J Component, ARVR course

MQTT telemetry with Unity AR overlay and ML-driven climate insights.

Tech: Unity, MQTT, AR, Python

<mailto:harishkumar56278@gmail.com?subject=AR%20Visualization%20Project>

Extractive Summarization & Information Retrieval

Where: JP Infotech

NLP web scraping with English-to-SQL query translation and 95% extraction accuracy.

Tech: Python, NLP, SQLite, MySQL

<mailto:harishkumar56278@gmail.com?subject=Summarization%20Project>

Application Tracking System

Where: JP Infotech

Resume analyser with OCR, keyword suggestions, and n-gram typing assist vs job descriptions.

Tech: Python, PyTesseract, OpenCV, NLP

<mailto:harishkumar56278@gmail.com?subject=Application%20Tracking%20System>

CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals

https://www.credly.com/badges/da3ef4b3-03a0-4560-95b6-5941473704de/public_url

- Applied Data Science (SmartInternz + Google Developers)

https://drive.google.com/file/d/126d8VTJ6y-X-t6_S_KOD2gYttWQhqe2T/view

- Data Analysis and Visualization with Power BI

<https://www.coursera.org/account/accomplishments/verify/YKAC25V5T6WJ>

- Preparing Data for Analysis with Microsoft Excel

<https://www.coursera.org/account/accomplishments/verify/F3SR649G78ZN>

- Harnessing the Power of Data with Power BI

<https://www.coursera.org/account/accomplishments/verify/CTOOZYTXL926>

- Extract, Transform, and Load Data in Power BI

<https://www.coursera.org/account/accomplishments/verify/5VXS5AHV12DN>